A web-based System Managing Course Files Using Google Platform: An Approach for Improving Academic Quality at a Higher Educational Institution

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Abstract –

Most of the higher educational institutions seeking academic accreditation of national and international agencies. These institutions seek accreditations for the graduate and undergraduate programs they offer. They record programs related academic activities, such as course assessments, assessment samples, grades, course specifications, and course reports in course files for quality improvement and future references. These institutions form committees and follow different approaches, e.g., Google platform to collect from faculty members and manage these files. This study aims to present an enhanced approach of managing academic course files on Google platform. The proposed approach is an effective way in minimizing the efforts and the associated cost. This study's method uses the university computing infrastructure, tools, and Google platform. The proposed approach applied for two consecutive semester and is produced efficient results. The approach is adoptable for any academic program in an institutional environment and open to redesign following the users' requirements for any educational discipline.

Keywords: Academic Course, Course Files, Google Platform, web-based System
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I. INTRODUCTION

Google forms is a web app and a member of the drive family of Docs, sheets, slides, and other apps [1]. This app enables a platform to conduct surveys in a quick way, get participants' responses, and publish them on the web. This app is significant in managing the survey's results, analyzing the results, and generating summary reports. Furthermore, the app enables an interface to the participants/users with several options, such as accessible in any browser and anywhere-anytime and supports social media sites [2]. Noticeably, Google forms freely available effective way to obtain responses, data, and information from the participants. Generally, the app obtains the participants' indented data, such as personal information, methods

of communication, and the event time or meeting time suitability. Equally, the owner who is conducting surveys also similar advantages, such as reports in different formats and the summary of the collected data [3]. Business establishments sue Google forms for marketing their products and getting feedback from their customers. These establishments use the summary reports provided by the app to improve their business performance. These organizations use the form for three reasons (i) easily accessible by any user, (ii) the app is free of cost, and (iii) it is a reliable interface [4].

Google forms are used for multiple purposes – for services in different business domains, organizations for their business activities, and academic institutions for most of their academic activities. Most of the academic institutions use Google forms for gathering students' data, getting feedback, and even it is used for short assessments. Besides, many academic institutions use Google forms for managing academic course files. A course file is a road map for the teaching process and a set of academic documents that are generated during a course delivery throughout the semester [5]. Furthermore, a course file consist of several items as listed in Table 1; these items generate by a course teacher over a period of course delivery and by the academic activities.

Its sources, means, methods of teaching, evaluation, outputs, and personal impressions of the one who teaches it. For example, course specification is one of the most important documents that includes, course associated entities, such as course learning outcomes, assessment methods, assessment types, learning resources, and so on.[6]

Most of the academic institutions follow the managing of course files on every academic year. Course teachers maintain the course files following the department guidelines. Table 1 lists the potential items of an academic course and the associated documents. The listed of items included in the course file are adopted following the guidelines on national academic quality control agencies, i.e., NCAAA. Remarkably, the presented study follows the agency's guidelines for managing academic course files. It uses Google forms for managing course files that can be more effective in an academic institutional environment[7].

Table 1: NCAAA course file items[8]

#s	Course file items and descriptions
1	Course specification-approved by course coordinator and HOD.
2	Course report-approved by course coordinator and HOD.
3	Course syllabus with weekly schedule including theory and lab/tutorial-approved by course coordinator and HOD
4	Copy of class and lab /tutorial attendance.
5	Copy of entered and confirmed marks from the registration system.
6	Copy of assessment final exam question paper-approved by course coordinator and HOD
7	Copy of final exam answer key-approved by course coordinator and HOD
8	Copy of mid exams question paper-approved by course coordinator and HOD
9	Copy of mid exams answer key-approved by course coordinator and HOD
10	Assessment samples of Mids. { question paper , answer } (in order)
11	Samples of final exam students answer paper [three samples - best, average and worst]
12	Assessment samples mid exams, quiz, assignment, lab mid exams and tutorials {three samples - best, average and worst} (in order).
13	Course materiel (power point & CD …..).

14	Cover of Text Book and List of Topics
15	Soft copies of course specification, course report, exams & course materiel saved in CD
16	Faculty CV
17	Semester timetable signed by HOD
18	CLOs matrix

II. LITERATURE REVIEW

A study [9] discussed various aspects about Google Forms in Education. It identified some tools for Google models, which can be used in the educational process to enhance communication between faculty members and students. Further, it highlighted that Google's models will be more effective for collecting and analyzing a large amounts of data. Furthermore, it focused the specific characteristic of Google Forms that facilitate collaborative work between members of a work team in the department in an organization environment.

Another study [10] emphasized the use and implementation of Google Forms for quality assurance systems and validating organizational administrative requirements. It highlighted the issue of quality learning in both formal and informal modes. Generally, an academic institution is very much concerned about quality standards and needs an effective system that facilitates academic activities and supports to produce essential documents and evidence required in an academic process. Indeed, the academic documents and evidence of an ongoing academic course in an institution need to be documented in a separate folder, which is called as course file. Every institution needs an effective mechanism for its educational process to manage the course file and required quality standards while managing such documents. The education process needs a strong quality system that can manage all the academic evidence that can be used for future purposes.

In literature, different types of systems were discussed, such as web-based system for course file management or course management system in an institutional facility [5][11]. These systems are effective in managing course files for the

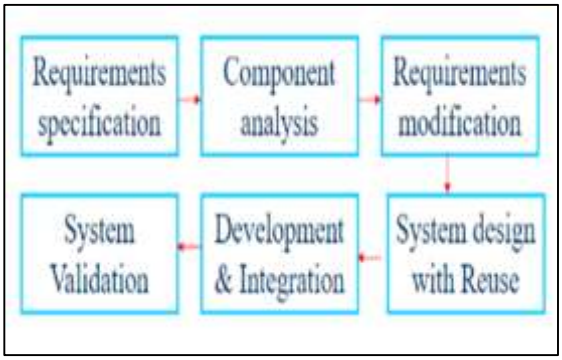
Current semester and for managing the previous semesters' repository. Such systems enable an easy to access and easy to manage platform for the administration for managing academic course files in an academic institutions.

III. METHODS

A. Reuse-oriented development model

This methodology depends on reusing the available software resources to design a reliable system, and characteristics of this methodology reducing the development cost of the system, as well as reducing the risks resulting from the development process of the systems[12][13]. Consists of four distinct phases: requirements planning, system design, development, and cutover [14]. In addition, the fast delivery of the finished product.

Figure1 Reuse-Oriented Development Model



B. Personal Interview

An information-gathering interview is a directed conversation with a specific purpose that uses a question-and-answer format.[15] The researchers focus in the personal interview is to be oriented around the status of the system, organizational and personal goals, and informal procedures for interacting with the system[16].

C. Interview Question

-What is your opinion of the current Mechanism of collecting the course file?

-Once the data are submitted via the paper file, how are they processed?

-Describe the monitoring process that is available on quality room that the specified to course files?

-What are some of the common errors made on receive course files?

-Do you support the idea of developing a mechanism for collecting course files through Google Forms?

D. Whom TO Interview

-University Vice-Dean for Development and Quality

-Dean for deanship of Development and Quality

-Vice-Dean for Development and Quality

-Heads of Development and Quality Units

E. Research Methodology Survey Questionnaire

In this study, we are using questionnaires as technique to information gathering that allows studying attitudes, beliefs, behavior,[17][18] and characteristics of several faculty members in the king Khalid University, who may be affected by the current and proposed systems.

A simple questionnaire was conducted on a group of faculty members, numbering about 80 faculty members, and the questionnaire was sent electronically through Google forms. Procedure for Paper Submission. As shown in table 3.

Table 2. Expert interview responses

Interview's participants / **Interviews' questions**	University deanship of academic development and quality	University vice deanship quality	College vice dean quality	Head of college vice dean quality
What's your opinion of the current Mechanism of collecting the course file	The current mechanism requires great effort and more time to collect, check and audit quality files for courses	The current mechanism requires a great effort and more time to collect, check and check the quality files of courses from	The current mechanism requires great effort and more time to collect, check and audit quality files for courses	All units in the faculties unanimously agreed that the current mechanism for collecting course files represents an additional burden for them at the end of

		members of the course files receiving team		each semester.
How the course file content is processed?	It is the responsibility of the development and quality unit in the college	It is the responsibility of the development and quality unit in the college	It is the responsibility of the development and quality unit in the college	It is ensured that all the required files are complete, and that the content of these files is complete, especially the course specification, the course report and the learning outcomes measurement matrix.
Describe the monitoring process that is available on quality room that the specified to course files?	It is the responsibility of the development and quality unit in the college	It is the responsibility of the development and quality unit in the college	It is the responsibility of the development and quality unit in the college	The process of monitoring and receiving course files is distributed among the members of the quality unit in the faculty so that members are allocated according to the specific academic department and the time available
What are some of the common errors made on receive course files?	Errors must be disclosed upon receiving the course file from the Quality Unit at the college, so as not to affect the preparation of future reports	Errors must be disclosed upon receiving the course file from the Quality Unit at the college, so as not to affect the preparation of future reports	Errors must be disclosed upon receiving the course file from the Quality Unit at the college, so as not to affect the preparation of future reports	- Some models used may not be current. - The content does not meet the requirements. - Some of the required documents may not be received
Do you support the idea of developing a mechanism for collecting course files through Google Forms?	Agrees to support	Agrees to support	Agrees to support	Agrees to support

Table 3. Survey questionnaire

Question	Strongly agree	Agree	Not sure	Disagree	Strongly disagree

1. Preparing the contents of the course file on a hard-copy indexed file requires time	60	10	5	3	2
2. Printing the contents of the course file on papers is expensive and requires effort and time	65	15	-	-	-
3. Do you support the use of technology to prepare and prepare the course file?	50	20	5	4	1
4. The process of delivering the traditional paper course file may take a longer time, especially if the person responsible for receiving is busy with other tasks	40	25	5	5	5
5. Do you support electronic submission of the course file without papers?	50	25	5	2	3
6. The process of checking and reviewing the course file sent electronically does not require time and effort	60	20	-	-	-
7. Modifying the content of the course file and re-delivering it electronically is considered easier and simpler than the traditional way	60	20	-	-	-

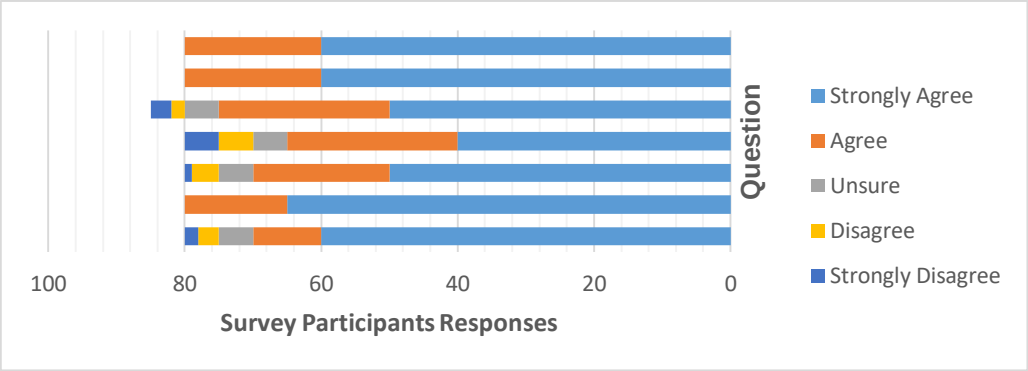


Figure 2 Survey questionnaire

VI. A Comparison of File Management Software Tools

The presented study has determine few file management systems that are in use for managing documents.

eFileCabinet: It is one of the easy to access document management systems that provides a robust document management solution with intelligent document tracking facility [19].

Microsoft OneDrive: It is an incredible documents and photos management facility provided by Microsoft. It offers a strong tools collaboration mechanism where users share their content in an easy way [20].

Dropbox: An innovative file management approach enables a common platform to store files, web shortcuts, and the cloud content [21].

Apple iCloud: It facilitates its users secure environment with numerous benefits, such as users can store all types of their content in the cloud. It manages the content intelligently with context-aware characteristics [22].

Google drive: It offers personal file management and easy access to personal content. It leverages intelligence from AI domain to offer its services much faster than the other ones [23].

DigitalDrawer: is an on premise file management system, which is more suitable for small and medium organizations. It manages content and made available for its users anywhere-anytime.
PinPoint: It is one of the most powerful document management platform and manages content digitally with a single location. It allows users to access all the facilities from any device and the users on move.

IV. RESULTS AND DISCUSSION

The purpose of the implemented approach to collect and manage academic course files for the quality development process in the department. It practices the file management following the university guidelines, which is the general practice in Saudi universities inline to National Qualification Framework. Google Drive includes the features that are feasible for the file management. The university made an agreement with the service provider for unlimited storage space and an additional feature of security, i.e., Google Authenticator. Therefore, this platform is most suitable for the course file management for the faculty members using their university email. Usually, a coordinator is assigned by the department to control and coordinate among the faculty members in the process.

F. Implementation result

For effective implementation, a few interfaces are designed to simulate course file management. Figures 3-6 illustrate the interfaces for different events in the course file management process. These figures describe the steps involve for collecting the files for the all the courses offered from the program curriculum on a semester.

To access this interface, the course coordinator must have a link to it, and then log in to via his account, which belongs to the institution and is hosted on Google services. And then write the name and press the Next button.

After filling the name in the first interface, it is moved to this interface, from which it is determined at which level the course is located, and then click on the Next button.

After selecting the level, all the courses it contains will appear according to the program's study plan. The course name is one of the most important data required, which will be linked to all files. So put in the design of user interface as an option, to prevent users from entering the course name incorrectly.

Figure 3 Step-1fill the nameof the coursecoordinator

Figure4. Step-2 Choose the course level
Figure5 Step-3 Choose the course

COLLEGE OF
BUSINESS
كلية الأعمال
Quality Course Files for MIS Dept. Courses
oanasar@kku.edu.sa Switch account
The name, email, and photo associated with your Google account will be recorded when you upload files and submit this form
* Required
Courses for level-3
Choose the course *
Introductions to Information Technology
Back
Next
Clear form
Never submit passwords through Google Forms.
Google Forms

Figure 6 Step-4 upload course files

After choosing the name of the course in the previous step, an interface will appear through which all files will be uploaded, knowing that the files are divided into two types (required and optional).

The question here is how to check and review the content of these files and confirm the validity of the data and information on them, and a mechanism has been developed for this, represented by:

- Assigning a coordinator for each program to review and audit course files and ensure their integrity and data validity.
- After that, the coordinator prepares a detailed report showing the missing elements for each course and the validity of the evidence and attached files.
- After that, the detailed report is submitted to the head of the program to circulate it to all those who teach the courses to correct the observations contained in the report, which may be correct data and files that were uploaded incorrectly.

G. Result analysis

The implement approach is an effective way to manage course files in an educational institution. The authors' of the study have implemented the systems for several semesters. The proper system of managing course files help the department to keep the

records efficiently. These records will useful for the department for the future requirements, such as program accreditation, university records, and for the national quality measure. Indeed, the implemented system should be an ideal model for any academic program irrespective of the academic disciplines.

The authors have conducted a survey to validate the study's claim. Table 4 describes the result of the survey, which shows 95% of the users were agreed the effectiveness of the system. Additionally, they were convinced the use of the system and the stored records would be useful for the future uses in different context.

Table 4. The survey's result analysis

	University deanship of academic development and quality		**University vice deanship quality**		**College vice dean quality**		**Head of college vice dean quality**		**Course coordinator**	
Number of users	Satisfied	Not satisfied	Satisfied	Not satisfied	Satisfied	Not satisfied	Satisfied	Not satisfied	Satisfied	Not satisfied
80	**1**	**0**	**2**	**0**	**4**	**0**	**6**	**0**	**60**	**7**

VII. CONCLUSION

The current study aims to provide an effective document and evidence management approach using Google platform, which is one of the significant contribution for managing academic activities, documents, and evidence in an institutional environment. On each semesters, teachers produce many academic documents, such as questions papers, assessment samples, and result summary. These documents must be properly manage by the teachers for various purposes (e.g., course improvement) and for the future use by the university administrations. Most of the institutions practice managing course files for multiple reasons, such as quality improvement, accreditations, and institutional ranking. The study's approach is one of the most effective way of managing academic documents in an institution. Teachers and institutions will find this approach very useful for managing academic documents, more precisely for an academic course file. This approach should be considered as a feasible model for managing academic documents for course file. Future studies should consider this model for adding more features, such as managing documentsbased on content similarities using machine learning algorithms.

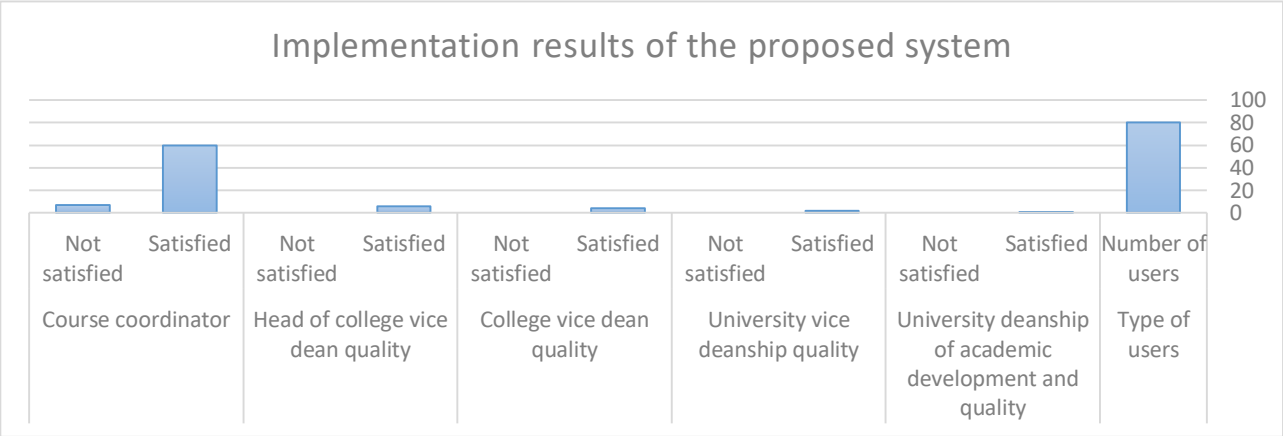


Figure 6 Implementation results of the proposed system

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